Synchrotron physics — Solution

Y26 (2026) — English translation

A1 — 0.50 pts

Determine the magnitude of the magnetic field $B(t)$ of the electromagnets at the moment when the energy of the electrons $E(t)$. Express your answer in terms of $E(t)$, T and physical constants.

Let's write down Newton's second law:

$$evB = p\frac{2\pi}{T}.$$

$$B = \frac{2\pi}{eT} \cdot \frac{p}{v} = \frac{2\pi}{eT} \cdot \gamma m = \frac{2\pi}{eT} \cdot \frac{E}{c^2}$$

Answer: \textbfAnswer:

$$B(t) = \frac{2\pi E(t)}{eTc^2}$$

A2 — 0.40 pts

Determine at what ratio there are equilibrium electrons on T, U and dB/dt.

Let's express the energy from the result of the previous point and differentiate taking into account the fact that $T = \text{const}$.

$$\frac{dE}{dt} = \frac{eTc^2}{2\pi} \cdot \frac{dB}{dt} = \frac{eU}{T}\cos\varphi$$

$$\cos\varphi = \frac{T^2c^2}{2\pi U} \cdot \frac{dB}{dt}$$

The cosine does not exceed 1, which provides a condition for the permissible parameters of the synchrotron.

Answer: \textbfAnswer:

$$\frac{T^2c^2}{2\pi U} \cdot \frac{dB}{dt} \leqslant 1$$

A3 — 0.50 pts

Find the dependence of frequency $\omega_U(t)$ on time. Express your answer in terms of $B(t)$, n, R and physical constants.

Newton's second law is similar to the previous points:

$$evB = p\frac{2\pi}{T} = \frac{p\omega_U}{n}.$$

Let us express p and E:

$$p = eB\frac{vT}{2\pi} = eBR.$$

$$E = \sqrt{(m_pc^2)^2 + (pc)^2} = \sqrt{(m_pc^2)^2 + (ceBR)^2}$$

Relation E and B можно погayandть and similarly прошлому party:

$$eB = \frac{p\omega_U}{vn} = \frac{E\omega_U}{c^2 n}.$$

$$\omega_U(t) = \frac{nec^2 B}{E}$$

All that remains is to substitute the energy value.

Answer: \textbfAnswer:

$$\omega_U(t) = \frac{nec^2 B(t)}{\sqrt{(m_p c^2)^2 + (ceB(t)R)^2}}$$

B1 0.50 pts

Obtain the value α for the weak focus case.

Similar to the previous point, the expression for momentum in a circular orbit with radius R:

$$p(R) = eB(R)R = \text{const} \cdot R^{1-\xi}.$$

Then relation на относительные изменения:

$$\frac{dp}{p} = (1-\xi)\frac{dR}{R} = (1-\xi)\frac{dL}{L}.$$

Answer: \textbfAnswer:

$$\alpha = \frac{1}{1-\xi}$$

B2 1.50 pts

Express K in terms of γ and α.

Let's use the expression for the cyclic frequency:

$$\omega = \frac{2\pi v}{L}.$$

$$\frac{d\omega}{\omega} = \frac{dv}{v} - \frac{dL}{L} = \frac{dv}{v} - \alpha\frac{dp}{p}$$

$$K = \frac{E}{v}\frac{dv}{dE} - \alpha\frac{E}{p}\frac{dp}{dE}$$

Let us find производные and substituting.

$$\frac{vdE}{Edv} = \frac{vd\frac{1}{\sqrt{1-v^2/c^2}}}{dv/\sqrt{1-v^2/c^2}} = -\frac{v}{\sqrt{1-v^2/c^2}} \cdot \frac{d\sqrt{1-v^2/c^2}}{dv} = \frac{v^2}{c^2-v^2} = \beta^2\gamma^2 = \left(1-\frac{1}{\gamma^2}\right)\gamma^2 = \gamma^2 - 1$$

$$\frac{p}{E}\frac{dE}{dp} = \frac{p}{\sqrt{(mc)^2+p^2}}\frac{d\sqrt{(mc)^2+p^2}}{dp} = \frac{p^2}{(mc)^2+p^2} = 1 - \frac{(mc^2)^2}{E^2} = 1 - \frac{1}{\gamma^2} = \frac{\gamma^2-1}{\gamma^2}$$

Answer: \textbfAnswer:

$$K = \frac{1-\alpha\gamma^2}{\gamma^2-1}$$

B3 — 0.50 pts

Note at what focusing the critical energy exists. Find γ for particles with critical energy.

For a critical situation we have:

$$\gamma = \frac{1}{\sqrt{\alpha}}.$$

$\gamma > 1$, therefore for the existence of critical energy it is necessary $\alpha < 1$. For weak focusing this condition is not met.

Answer: \textbfAnswer: Critical energy only exists for strong focus.

$$\gamma = \frac{1}{\sqrt{\alpha}}$$

C1 — 0.90 pts

Considering that the initial energies of the particles differ only slightly, express $\dot{\varphi}$ in terms of K, $\mathcal{E}$, n, ω_0 and E_0.

Let us consider the change in the phase of a particle in contrast to the equilibrium one during one revolution:

$$\Delta\varphi = \omega_U \left(\frac{2\pi}{\omega} - \frac{2\pi}{\omega_0}\right).$$

Dividing на time этого изменения, that is period обращения частицы equalй $2\pi/\omega$.

$$\dot{\varphi} = \omega_U \left(1 - \frac{\omega}{\omega_0}\right) = n(\omega_0 - \omega)$$

Let us use olimiteнием K and we get:

$$\omega - \omega_0 = K\frac{\omega_0}{E_0}\mathcal{E}.$$

Answer: \textbfAnswer:

$$\dot{\varphi} = -\frac{nK\omega_0}{E_0}\mathcal{E}$$

C2 — 0.30 pts

Express $\dot{\mathcal{E}}$ in terms of φ_0, φ, U, ω_0 and physical constants. Use the fact that the difference between ω and ω_0 is small enough to be neglected in this case.

Let us write down the derivatives E and E_0.

$$\dot{E} = \frac{\omega}{2\pi}eU\cos\varphi \approx \frac{\omega_0}{2\pi}eU\cos\varphi$$

$$\dot{E}_0 = \frac{\omega_0}{2\pi}eU\cos\varphi_0$$

Answer: \textbfAnswer:

$$\dot{\mathcal{E}} = \frac{\omega_0 eU}{2\pi}\left(\cos\varphi - \cos\varphi_0\right)$$

C3 — 0.60 pts

Obtain an expression for $\ddot{\varphi}$ in terms of K, E_0, ω_0, U, φ_0, φ, n and physical constants. Determine at what φ_0 the equilibrium orbit is stable (that is, the initially infinitesimal deviations φ do not grow with time).

Let us differentiate $\dot{\varphi}$ taking into account the indicated approximations.

$$\ddot{\varphi} = -\frac{nK\omega_0}{E_0}\dot{\mathcal{E}}$$

For stability, we expand to first order:

$$\ddot{\varphi} = \frac{nK\omega_0^2 eU}{2\pi E_0}\sin\varphi_0(\varphi - \varphi_0).$$

Answer: \textbfAnswer:

$$\ddot{\varphi} = -\frac{nK\omega_0^2 eU}{2\pi E_0}\left(\cos\varphi - \cos\varphi_0\right)$$

C4 — 0.50 pts

Express $\dot{\varphi}^2$ in terms of K, E_0, ω_0, U, φ_0, φ, n, φ_i and $\mathcal{E}_i$ and physical constants.

Let's use the result of the previous paragraph.

$$\ddot{\varphi} = \frac{d\dot{\varphi}}{dt} = \frac{d\dot{\varphi}}{d\varphi}\frac{d\varphi}{dt} = \frac{d\dot{\varphi}^2}{2d\varphi}$$

$$d\dot{\varphi}^2 = -\frac{nK\omega_0^2 eU}{\pi E_0}\left(\cos\varphi - \cos\varphi_0\right)d\varphi$$

$$\dot{\varphi}^2 + \frac{nK\omega_0^2 eU}{\pi E_0}\left(\sin\varphi - \varphi\cos\varphi_0\right) = \text{const}$$

Осталось подставить начальные значения.

$$\dot{\varphi}_i = -\frac{nK\omega_0}{E_0}\mathcal{E}_i$$

Answer: \textbfAnswer:

$$\dot{\varphi}^2 = \left(\frac{nK\omega_0}{E_0}\mathcal{E}_i\right)^2 + \frac{nK\omega_0^2 eU}{\pi E_0}\left(\sin\varphi_i - \varphi_i\cos\varphi_0\right) - \frac{nK\omega_0^2 eU}{\pi E_0}\left(\sin\varphi - \varphi\cos\varphi_0\right)$$

C5 — 1.80 pts

Determine at what values of φ_i and $\mathcal{E}_i$ the particle is involved in the acceleration process, that is, on average it receives the same amount of energy as the equilibrium one. Qualitatively depict the boundaries of the found region of the $(\varphi_i, \mathcal{E}_i)$ plane for $\varphi_0 = 0$ (resonant acceleration) and $\varphi = \pi/4$.

Let's use the result of the previous point:

$$\dot{\varphi}^2 + \frac{nK\omega_0^2 eU}{\pi E_0}\left(\sin\varphi - \varphi\cos\varphi_0\right) = \text{const} = \left(\frac{nK\omega_0}{E_0}\mathcal{E}_i\right)^2 + \frac{nK\omega_0^2 eU}{\pi E_0}\left(\sin\varphi_i - \varphi_i\cos\varphi_0\right).$$

Note that вовлечение в acceleration equalsсильно ограниченности движения φ.

$$\dot{\varphi}^2 + V(\varphi) = \text{const} = H$$

Погayandли задачу исследования движения в некотором "potentiale" V с "энергией" H. Consider сгуай $K > 0$. Построим качественный вид V for this case.

Red indicates the maximum level of "energy" at which the movement is finite (occurs in a limited area). With greater "energy" φ can grow unlimitedly (movement to the right on the graph). Let's find this value H. It is determined by the value of the maximum V.

$$\frac{dV}{d\varphi} = \frac{nK\omega_0^2 eU}{\pi E_0}\left(\cos\varphi - \cos\varphi_0\right)$$

Максимум соответствует $\varphi = |\varphi_0|$.

$$H = \frac{nK\omega_0^2 eU}{\pi E_0} \left(\sin|\varphi_0| - |\varphi_0|\cos\varphi_0\right)$$

Then condition на φ_i and $\mathcal{E}_i$ имеет вид:

$$\left(\frac{nK\omega_0}{E_0}\mathcal{E}_i\right)^2 + \frac{nK\omega_0^2 eU}{\pi E_0}\left(\sin\varphi_i - \varphi_i\cos\varphi_0\right) \leqslant \frac{nK\omega_0^2 eU}{\pi E_0}\left(\sin|\varphi_0| - |\varphi_0|\cos\varphi_0\right).$$

Стоит отметить, что at the same time существует дополнительное condition на φ_i. Fromначально частица должна находится в "potentialьной яме", что соответствует $\varphi_i \leqslant |\varphi_0|$. Note that for/at $K < 0$ ситуация аналогичная, но теперь potential имеет максимум for/at $\varphi = -|\varphi_0|$. Then максимальная "energy":

$$H = \frac{nK\omega_0^2 eU}{\pi E_0}\left(-\sin|\varphi_0| + |\varphi_0|\cos\varphi_0\right).$$

И similarlye ограничение на фазу $\varphi_i \geqslant -|\varphi_0|$. Note that можно обобщить на произвольный sign K:

$$H = \frac{n|K|\omega_0^2 eU}{\pi E_0}\left(\sin|\varphi_0| - |\varphi_0|\cos\varphi_0\right).$$

$$\frac{K}{|K|}\varphi_i \leqslant |\varphi_0|$$

Final expression for the desired area:

Region border:

$$\left(\frac{nK\omega_0}{E_0}\mathcal{E}_i\right)^2 + \frac{nK\omega_0^2 eU}{\pi E_0}\left(\sin\varphi_i - \varphi_i\cos\varphi_0\right) = \frac{n|K|\omega_0^2 eU}{\pi E_0}\left(\sin|\varphi_0| - |\varphi_0|\cos\varphi_0\right);$$

$$\frac{K}{|K|}\varphi_i \leqslant |\varphi_0|.$$

Case $\varphi_0 = 0$

$$\left(\frac{nK\omega_0}{E_0}\mathcal{E}_i\right)^2 + \frac{nK\omega_0^2 eU}{\pi E_0}\left(\sin\varphi_i - \varphi_i\right) = 0;$$

$$K\varphi_i \leqslant 0.$$

Несложно заметить, что for/at $\varphi_i \neq 0$:

$$K\left(\sin\varphi_i - \varphi_i\right) = -K\varphi_i\left(1 - \frac{\sin\varphi_i}{\varphi_i}\right).$$

At the same time $1 - \frac{\sin\varphi_i}{\varphi_i} > 0$, therefore:

$$K\left(\sin\varphi_i - \varphi_i\right) = -K\varphi_i\left(1 - \frac{\sin\varphi_i}{\varphi_i}\right) > 0.$$

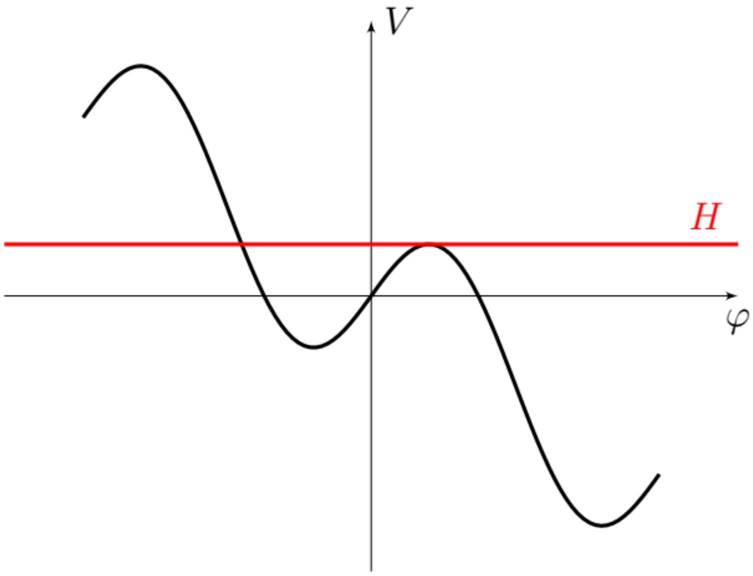
Substituting в первое нeequality and we get:

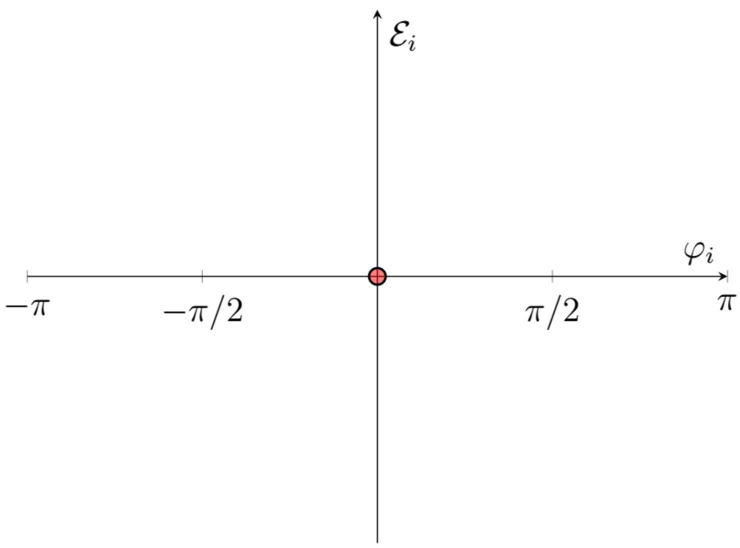
$$\left(\frac{nK\omega_0}{E_0}\mathcal{E}_i\right)^2 < 0.$$

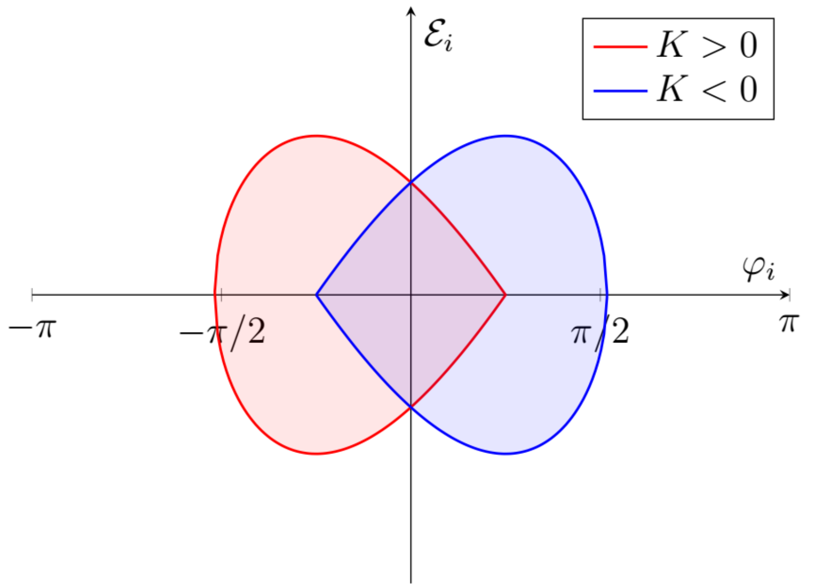
Thus, единственная точка, удовлетворяющая условиям $(\varphi_i, \mathcal{E}_i) = (0, 0)$. Стоить отметить, что данный факт очевиден from рассмотрения энергетической аналогии. For/At $\varphi_0 = 0$ potential has no minimum, there is only an inflection point at zero, so the only case of limited movement is equilibrium at zero.

Case $\varphi_0 = \pi/4$

$$\left(\frac{nK\omega_0}{E_0}\mathcal{E}_i\right)^2+\frac{nK\omega_0^2eU}{\pi E_0}\left(\sin\varphi_i-\frac{\varphi_i}{\sqrt{2}}\right)=\frac{n|K|\omega_0^2eU}{\pi E_0}\left(\frac{1}{\sqrt{2}}-\frac{\pi}{4\sqrt{2}}\right)$$
$$\frac{K}{|K|}\varphi_i\leqslant\frac{\pi}{4}$$
Погayandли два cгayaя в зависимости от signa K:
Answer: \textbfAnswer:
$$\left(\frac{nK\omega_0}{E_0}\mathcal{E}_i\right)^2+\frac{nK\omega_0^2eU}{\pi E_0}\left(\sin\varphi_i-\varphi_i\cos\varphi_0\right)\leqslant\frac{n|K|\omega_0^2eU}{\pi E_0}\left(\sin|\varphi_0|-|\varphi_0|\cos\varphi_0\right);$$
$$\frac{K}{|K|}\varphi_i\leqslant|\varphi_0|.$$






$\mathcal{E}_i$
$K > 0$
$K < 0$
φ_i
$-\pi$
$-\pi/2$
$\pi/2$
π